DIFFUSION ON SG(2)

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Contents

I follow mainly [Bar98, Section 2].

1. Notation

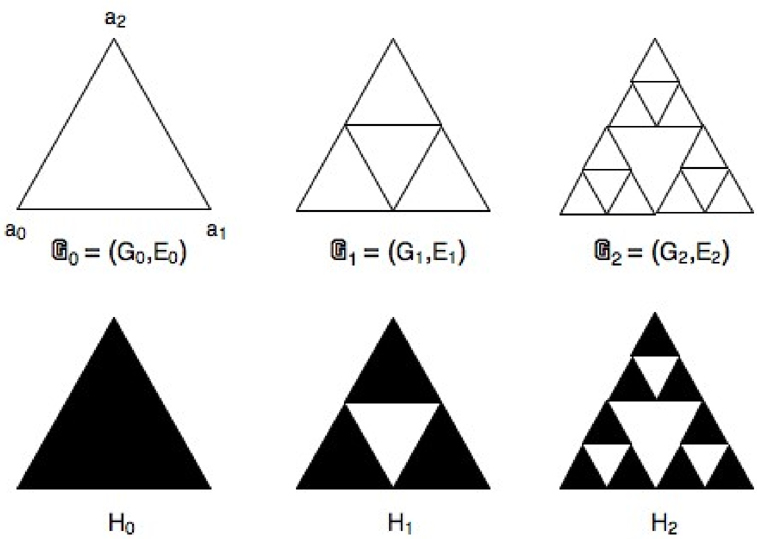

Figure 1. SG(2) approximating graphs and sets.

H_0 is the closed convex unit triangle in $\mathbb{R}^2$. $G_0 = \{a_1, a_2, a_3\}$ is the set of vertices.
H_n is the union of 3^n scaled copies of H_0, each of side 2^{-n}. G_n is the set of vertices of H_n.
E_n is the set of edges. $\mathbb{G}_n := (G_n, E_n)$ is the corresponding graph.

$G = SG(2) := \bigcap_{n\geq 0} H_n$.

An *n-triangle* is any one of the 3^n triangles in H_n.

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An n-*complex* is a set of the form $G \cap B$ where B is an n-triangle.

$\mathcal{S}_n$ is the set of 3^n n-complexes.

For $x \in G_n$, $D_n(x) = \bigcup\{S \in \mathcal{S}_n : x \in S\}$. (So unless $x \in \{a_0, a_1, a_2\}$, there are exactly two sets in this union.)

2. Flat Measure

Define μ_n to be Lebesgue measure restricted to H_n, normalised so $\mu_n(H_n) = 1$.

Define the *flat measure* on G by $\mu_G :=$ weak $\lim_{n\to\infty}\mu_n$. It assigns mass 3^{-n} to each n-cell.

Define the *fractal dimension* $d_f := \log 3/\log 2$. It is the scaling dimension for fractals generated by similitudes and satisfying the open set condition. It is also the Hausdorff dimension.

It follows (easy) for $x \in G$ and $0 \le r < 1$ that

$$3^{-1}r^{d_f} \le \mu_G B_r(x) \le 18r^{d_f}.$$

(The constants 3 and 18 are unimportant.)

3. Geodesic Metric

For any *closed* $A \subset \mathbb{R}^d$ and $x, y \in A$ define

$$d_A(x,y) = \inf\{|\gamma| : \gamma \text{ is a path between } x \text{ and } y \text{ with } \gamma \subset A\}. \tag{1}$$

If d_A is finite for all $x, y \in A$ then d_A is a metric. It is called the *geodesic metric* on A. In this case, (A, d_A) satisfies the geodesic property:

For each $x, y \in A$ there exists a map $\Phi : [0,1] \to A$ such that

$$d_A(x,\Phi(t)) = t d_A(x,y), \quad d_A(\Phi(t),y) = (1-t)d_A(x,y), \quad \forall t \in [0,1].$$

Proof: Take a sequence of approximating maps and use a diagonal argument.

For G one obtains

$$|x-y| \le d_G(x,y) \le 8|x-y|. \tag{2}$$

Proof: By a chaining argument.

(As usual, the constants 1 and 8 are unimportant.)

4. SRW on the prefractals

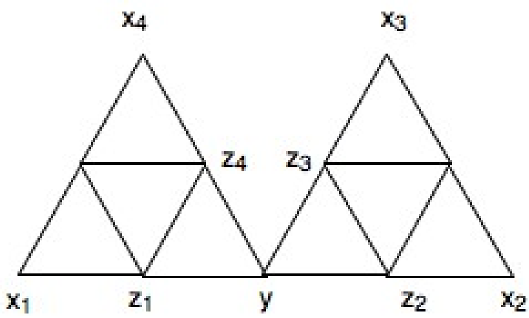

FIGURE 2. A subgraph of $\mathbb{G}_n$. $y \in \mathbb{G}_{n-1}$ and its neighbours $x_1, \dots, x_4 \in \mathbb{G}_{n-1}$. These 5 points and the remaining 6 nodes are all in $\mathbb{G}_n$.

4.1. Construction.

Let $(Y_k^{(n)})_{k\ge 0}$ be the simple random walk [SRW] on $\mathbb{G}_n$. Unless otherwise indicated, $Y_0^{(n)} = a_0 = 0$, see Figure 1.

In Figure 2

$$P(Y_{k+1}^{(n)} = z_i \mid Y_k^{(n)}) = y) = \frac{1}{4}, \quad P(Y_{k+1}^{(n-1)} = x_i \mid Y_k^{(n-1)}) = y) = \frac{1}{4}.$$

Let $0 = T_0, T_1, T_2, \dots$ denote successive disjoint[1] hits by $Y_k^{(n)}$ on $\mathbb{G}_{n-1}$. Then by symmetry, restricting $Y_k^{(n)}$ to successive hits on $\mathbb{G}_{n-1}$, we get

$$Y_{T_k}^{(n)} \stackrel{d}{=} Y_k^{(n-1)} \quad k \geq 0.$$

That is, the restricted SRW on the left is a representative of the SRW on the right.

Fixing n and restricting to $m < n$ we get a sequence of nested SRW's on $\mathbb{G}_m$ for each $m \leq n$. Moreover, we can similarly nest $Y^{(n)}$ in $Y^{(n+1)}$ by building in a sequence of independent excursions between $Y_k^{(n)}$ and $Y_{k+1}^{(n)}$.[2] (See [Bar98, p15] for a brief discussion and the fact that this can also be done by using the Kolmogorov extension theorem.) In this manner we get a prob space $(\Omega, \mathcal{F}, \mathbb{P})$ and random variables $(Y_k^{(n)})_{k \geq 0}$ for $n \geq 0$ s.t.

(1) $(Y_k^{(n)})_{k \geq 0}$ is a SRW on $\mathbb{G}_n$ beginning at $a_0 = 0$,

(2) If $m < n$ and $0 = T_0^{m,n}, T_1^{m,n}, T_2^{m,n}, \dots$ are successive disjoint hits by $(Y_i^{(n)})_{i \geq 0}$ on $\mathbb{G}_m$, then

$$Y_{T_k^{m,n}}^{(n)} = Y_k^{(m)} \quad k \geq 0, \tag{3}$$

with equality *everywhere*, not just in distribution.

Note

(1) The right side of (3) is independent of $n > m$.

(2) For $T_k^{m,n} \leq i \leq T_{k+1}^{m,n}$, $|Y_i^{(n)} - Y_k^{(m)}| \leq 2^{-m}$, since $Y_i^{(n)}$ is in the union of the two m-cells containing $Y_k^{(m)}$.

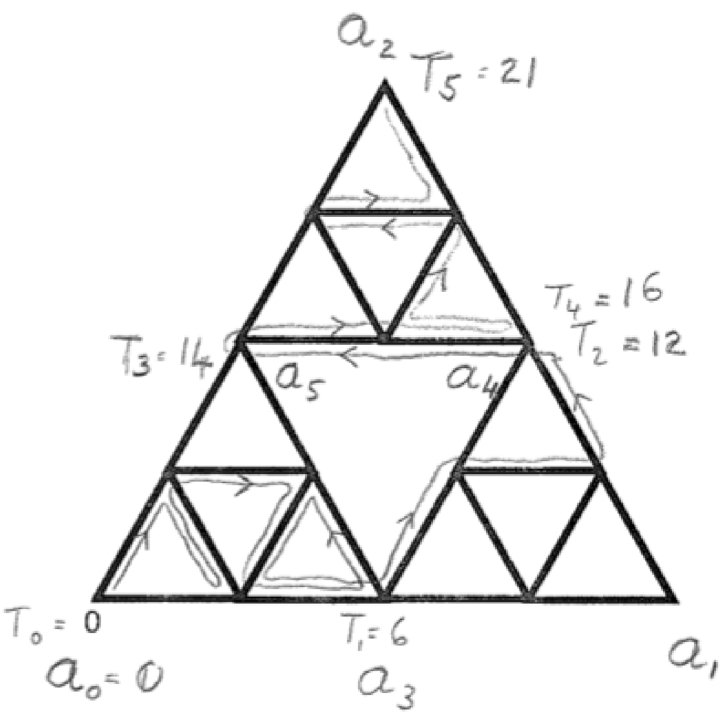

FIGURE 3. A sample path for $(Y_k^{(2)})_{k \geq 0}$ on $\mathbb{G}_2$, and the nested walks $(Y_k^{(1)})_{k \geq 0}$ on $\mathbb{G}_1$ and $(Y_k^{(0)})_{k \geq 0}$ on $\mathbb{G}_0$.

[1] That is, $Y_{T_{i+1}}^{(n)} \neq Y_{T_i}^{(n)}$.

[2] Note the following. In Figure 3 the path for $Y^{(1)}$ is $a_0 a_3 a_4 a_5 a_6 a_2$ but its extension $Y^{(2)}$ goes outside the 1-cell determined by $a_3 a_4$ in its journey from a_3 to a_4. However, it *cannot* in its journey from a_3 to a_4 go outside the *two* 1-cells containing a_3.

Note also that the crossing time of $Y^{(2)}$ from a_3 to a_4 is the same as if it were restricted to the 1-cell $a_3 a_1 a_4$. This follows from a simple reflection argument at the point a_3.

Example: In Figure 3 we show a sample path for $(Y_k^{(2)})_{k\geq 0}$ on $\mathbb{G}_2$. The *successive disjoint hits* on $\mathbb{G}_1$ are at times

$$T_0 = T_0^{2,1} = 0, \qquad T_1 = T_1^{2,1} = 6, \qquad T_2 = T_2^{2,1} = 12,$$
$$T_3 = T_3^{2,1} = 14, \qquad T_4 = T_4^{2,1} = 16, \qquad T_5 = T_5^{2,1} = 21.$$

The induced nested process $(Y_k^{(1)})_{k\geq 0}$ on $\mathbb{G}_1$ is

$$Y_0^{(1)} = Y_0^{(2)} = a_0, \qquad Y_1^{(1)} = Y_6^{(2)} = a_3, \qquad Y_2^{(1)} = Y_{12}^{(2)} = a_4$$
$$Y_3^{(1)} = Y_{14}^{(2)} = a_5, \qquad Y_4^{(1)} = Y_{16}^{(2)} = a_4, \qquad Y_5^{(1)} = Y_{21}^{(2)} = a_2.$$

The *successive disjoint hits* of $(Y_k^{(2)})_{k\geq 0}$ and $(Y_k^{(1)})_{k\geq 0}$ on $\mathbb{G}_0$ are at times

$$T_0^{2,0} = T_0^{1,0} = 0, \qquad T_1^{2,0} = 21, \qquad T_1^{1,0} = 5.$$

The induced nested process $(Y_k^{(0)})_{k\geq 0}$ on $\mathbb{G}_0$ is

$$Y_0^{(0)} = Y_0^{(1)} = Y_0^{(2)} = a_0, \qquad Y_1^{(0)} = Y_5^{(1)} = Y_{21}^{(2)} = a_2.$$

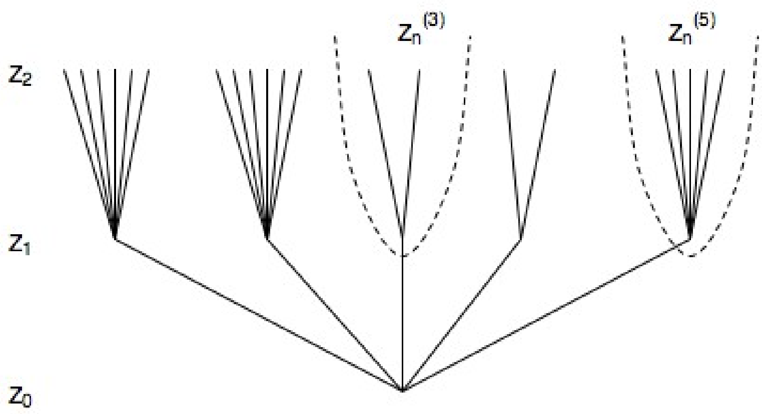

FIGURE 4. Crossing time branching process and sub processes.

(See Figures 3, 4 and Section 4.2.) The *branching process* of crossing times Z_n for $Y^{(n)}$ from $0 = a_0$ to $\{a_1, a_2\}$ here has values

$$Z_0 = 1, \quad Z_1 = 5, \quad Z_2 = 21.$$

The branching (sub-) processes $Z_n^{(i)}$ of crossing times between the $i-1$th and ith successive hit for $Y^{(n)}$ on $\mathbb{G}_1$ have values

$$Z_1^{(1)} = 1, \quad Z_1^{(2)} = 1, \quad Z_1^{(3)} = 1, \quad Z_1^{(4)} = 1, \quad Z_1^{(5)} = 1,$$
$$Z_2^{(1)} = 6, \quad Z_2^{(2)} = 6, \quad Z_2^{(3)} = 2, \quad Z_2^{(4)} = 2, \quad Z_2^{(5)} = 5.$$

4.2. **The branching process** Z_n**.** Let Z_n be the crossing time for $(Y_k^{(1)})_{k\geq 0}$ from $0 = a_0$ to $\{a_1, a_2\}$ in Figures 1 and 2. Let $\tau = Z_1$ $(= T_1^{0,1})$.

Then by construction $(Z_n)_{n\geq 0}$ is a simple branching process with

$$Z_0 = 1, \quad Z_{n+1} = \sum_{i=1}^{Z_n} Z_{1,i} = \sum_{j=1}^{Z_1} Z_{n,j}, \tag{4}$$

with iid $Z_{1,i} \overset{d}{=} \tau$ giving the offspring distribution and iid $Z_{n,j} \overset{d}{=} Z_n$.

The first equality is clear and the second is obtained by taking the Z_1 trees rooted at level one.

4.3. **The p.g.f. for Z_n.** In order to investigate Z_n it is natural to find its *probability generating function* [p.g.f.].[3] Consider first $\tau = Z_1$:

$$f(s) := \mathbb{E}(s^\tau) = \sum_{k\geq 0} P(\tau = k)s^k. \tag{5}$$

To compute this, let

$$\begin{aligned} f_c(s) &= \mathbb{E}^c(s^\tau) \quad \text{where } c \text{ is the midpoint of } 0a_1 \text{ or } 0a_2, \\ f_b(s) &= \mathbb{E}^b(s^\tau) \quad \text{where } b \text{ is the midpoint of } a_1a_2. \end{aligned}$$

Then it is easy to see from Figure 1 that

$$\begin{aligned} f(s) &= sf_c(s) \\ f_c(s) &= \frac{1}{4}s(f(s) + f_c(s) + f_b(s) + 1) \\ f_b(s) &= \frac{1}{2}(f_c(s) + 1). \end{aligned}$$

Solving,

$$f(s) = s^2/(4 - 3s), \quad \mathbb{E}\tau = f'(1) = 5, \quad \mathbb{E}\tau^k < \infty\ \forall k. \tag{6}$$

More generally, let

$$f_n(s) = \mathbb{E}s^{Z_n} = \sum_{k\geq 0} P(Z_n = k)s^k.$$

Then

$$f_0(s) = s, \quad f_1(s) = f(s) = \frac{s^2}{4-3s}, \quad f_n(s) = \overset{1}{f} \circ \cdots \circ \overset{n}{f}(s). \tag{7}$$

The first is immediate since $Z_0 = 1$ and the second is from (6). The third is a standard branching process / p.g.f. argument using either the fact that Z_{n+1} is Z_n iid copies of $\tau \overset{d}{=} Z_1$, or is Z_1 iid copies of Z_n (consider the trees rooted at level one).[4]

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[3] *Properties of a p.g.f.*: (see [Lal07]). If Z is a non negative integer valued R.V. then its p.g.f. is defined by

$$f(s) = \sum_{k\geq 0} P(Z = k)s^k = \mathbb{E}(s^Z).$$

The power series for f converges absolutely for $|s| < 1$, and the coefficients contain all the probabilities needed to describe Z.

Clearly,

$$\begin{aligned} f'(s) &= \sum_{k\geq 0} kP(Z = k)s^{k-1} = \mathbb{E}(Zs^{Z-1}), \\ f''(s) &= \sum_{k\geq 0} k(k-1)P(Z = k)s^{k-2} = \mathbb{E}(Z(Z-1)s^{Z-2}). \end{aligned}$$

All moments can be computed from the derivatives of f at 1. In particular,

$$f(1) = 1, \quad f'(1) = \mathbb{E}(Z), \quad f''(1) = \mathbb{E}(Z^2) - \mathbb{E}(Z),$$

and so

$$\operatorname{Var} Z = \mathbb{E}(Z - \mathbb{E}Z)^2 = f''(1) - f'(1) - (f'(1))^2.$$

[4] More precisely

$$\begin{aligned} f_{n+1}(s) &= \mathbb{E}s^{Z_{n+1}} = \mathbb{E}s^{\sum_{i=1}^{Z_1} Z_{n,i}} \quad \text{for iid } Z_{n,i} \overset{d}{=} Z_n \\ &= \mathbb{E}\prod_{i=1}^{Z_1} s^{Z_{n,i}} = \mathbb{E}f_n(s)^{Z_1} \quad \text{since } \mathbb{E}s^{Z_{n,i}} = f_n(s) \\ &= f(f_n(s)). \end{aligned}$$

Or

$$\begin{aligned} f_{n+1}(s) &= \mathbb{E}s^{Z_{n+1}} = \mathbb{E}s^{\sum_{j=1}^{Z_n} Z_{1,j}} \quad \text{for iid } Z_{1,j} \overset{d}{=} Z_1 \\ &= \sum_{k\geq 0} P(Z_n = k)\mathbb{E}s^{\sum_{j=1}^{k} Z_{1,j}} = \sum_{k\geq 0} P(Z_n = k)f(s)^k \\ &= \mathbb{E}f(s)^{Z_n} = f_n(f(s)). \end{aligned}$$

4.4. **Rescaled process W_n and its limit.** Since

$$Z_{n+1} = \sum_{i=1}^{Z_n} Z_{1,i} \quad \text{for iid } Z_{1,i} \overset{d}{=} Z_1,$$

it follows

$$\mathbb{E}(Z_{n+1} \mid Z_n) = (\mathbb{E}Z_1)\, Z_n = 5Z_n$$

Let $W_n = Z_n/5^n$. Then

$$\mathbb{E}(W_{n+1} \mid W_n) = W_n, \quad \mathbb{E}\mathbb{W}_{\Bbbk}{}^2 < \infty,$$

and so by martingale theory, see [Har63],[5]

$$Z_n/5^n = W_n \to W \text{ a.s.,} \tag{8}$$

where $0 < W < \infty$ a.s.

In particular, the growth rate of Z_n is of order 5^n.

4.5. **Subtrees $Z_n^{(i)}$ of Z_n.** The process Z_n gives the number of steps required for $Y^{(n)}$ to cross $\mathbb{G}_0$. After dividing by 5^{-n} we get a.s. convergence to a (continuous) positive finite distribution W.

Fix m and consider those $n \geq m$. Let $Z_{n,m}^{(i)}$ denote the number of steps taken by $Y^{(n)}$between hit $i-1$ and hit i on $\mathbb{G}_m$. That is

$$Z_{n,m}^{(i)} = T_i^{n,m} - T_{i-1}^{n,m}. \tag{9}$$

These are independent for fixed n and m, and on normalising we get

$$W_m^{(i)} := 5^{-n} Z_{n,m}^{(i)} \to W_m^{(i)} \text{ (say) } \overset{d}{=} 5^{-m} W, \text{ a.s. as } n \to \infty, \tag{10}$$

where for each m the $W_m^{(i)}$ are independent.

5. Construction of Diffusion Process on SG(2)

Let

$$X_t^n = Y_{[5^n t]}^{(n)} \quad t \geq 0, \tag{11}$$

where "$[\cdot]$ " is the "integer part of". This is the natural time scaling.

Theorem 5.1. *The process X_t^n converges a.s. and uniformly on compact intervals to a continuous process $(X_t)_{t\geq 0} \in G$.*

Proof Outline. Fix $m \geq 0$ and $i \geq 0$.

Suppose

$$\sum_{j=1}^{i} W_j^{(m)} < t < \sum_{j=1}^{i+1} W_j^{(m)}$$

(The idea is that t will be in the time interval between successive hits i and $i+1$ on $\mathbb{G}_m$ of the limit process X_t.)

Then from (9) and (10) for $n \geq n_0(\omega)$, say,

$$5^{-n} T_i^{m,n} < t < 5^{-n} T_{i+1}^{m,n}.$$

It follows from Note(2) following (3) that

$$|X_t^n - X_t^{n'}| \leq 2^{-m+1}$$

for $n, n' \geq n_0$.

Thus X_t^n is Cauchy in n and so converges a.s. for each t.

The fact convergence is uniform on compact intervals and the limit is continuous is by a similar argument. See [Bar98] for details. ☐

[5] *** Include some info on branching processes and this result

6. Properties of Diffusion Process on SG(2)

Theorem 6.1.

(1) *There exists a continuous strong Markov*[6] *process* $X = (X_t, t \geq 0, P^x, x \in G)$ *on* G.

(2) *The semigroup on* $\mathcal{C}(G)$ *defined by*

$$P_t f(x) = \mathbb{E} f(X_t)$$

is Feller[7] *and is* μ_G *symmetric:*

$$\int_G f\, P_t g\, d\mu_G = \int_G P_t f\, g\, d\mu_G.$$

(3) X *is locally isotropic*[8] *w.r.t. the metric* ${}_G$ *and the Euclidean metric.*

(4) *Let* $(T_i^n)_{i \geq 0}$ *denote successive disjoint hits by* X_t *on* $\mathbb{G}_n$. *Then* $\widehat{Y}_i^{(n)} := X_{T_i^n}$ *defines a SRW on* $\mathbb{G}_n$. *Moreover,* $\widehat{Y}_{[5^n t]}^{(n)} \to X_t$ *uniformly on compact sets a.s.*

In particular, X_t *is the process in Section 5.*

Proof. See [Bar98, Sections 6,7]. The main point is to prove (1). □

References

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[Lal07] Steven Lalley, *Generating function methods* (2007), available at `galton.uchicago.edu/~lalley/Courses/313/GeneratingFunctions.pdf`. lecture course notes.

[Lam77] John Lamperti, *Stochastic processes*, Springer-Verlag, 1977. A survey of the mathematical theory; Applied Mathematical Sciences, Vol. 23.

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[6] See [Lam77, Chapter 9].) The strong Markov property is that a.s.

$$P^x(X_{\tau+t} \in D \mid \mathcal{F}_\tau) = P^{X_\tau}(X_t \in D),$$

for any stopping time τ and associated σ-algebra $\mathcal{F}_\tau$.

[7] That is, $P_t f$ i continuous if f is continuous.

[8] If $\phi : A \to B$ and $\phi : \partial A \to \partial B$ are isometries where $A, B \subset G$, then

$$\forall x \in A\ \forall E \subset A \quad P^x(X_{t \wedge T_{\partial A}} \in E,\ t \geq 0) = P^{\phi(x)}(X_{t \wedge T_{\partial B}} \in \phi(E),\ t \geq 0).$$

Here $t \wedge T_{\partial A}$ is the minimum of the time t and the time $T_{\partial A}$ of first reaching the boundary ∂A. Similarly for $t \wedge T_{\partial B}$.